

SIEMENS



CloudLink

User Guide

Edition notice

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Table of Contents

	Edition notice.....	000
1	Cyber security disclaimer.....	5
2	Installation	6
3	CloudLink.....	8

1 Cyber security disclaimer

Siemens provides a portfolio of products, solutions, systems and services that includes security functions that support the secure operation of plants, systems, machines and networks. In the field of Building Technologies, this includes building automation and control, fire safety, security management as well as physical security systems.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art security concept. Siemens' portfolio only forms one element of such a concept.

You are responsible for preventing unauthorized access to your plants, systems, machines and networks which should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place. Additionally, Siemens' guidance on appropriate security measures should be taken into account. For additional information, please contact your Siemens sales representative or visit the following website:

<https://www.siemens.com/global/en/products/automation/topic-areas/industrial-cybersecurity.html>.

Siemens' portfolio undergoes continuous development to make it more secure. Siemens strongly recommends that updates are applied as soon as they are available and that the latest versions are used. Use of versions that are no longer supported, and failure to apply the latest updates may increase your exposure to cyber threats. Siemens strongly recommends to comply with security advisories on the latest security threats, patches and other related measures, published, among others, under the following website:

<https://www.siemens.com/cert/> => 'Siemens Security Advisories'.

2 Installation

The CloudLink installer software is a Siemens self-extracting performable application that installs the CloudLink software on your host computer. The CloudLink is a non-licensed tool to connect devices to Building X.

Prerequisites

- Verify that your current Windows version is up to date.
- Microsoft .NET Framework 8.0 and ASP.NET 8.0 must be available on the machine.
- Requires a Building X account and Building X access.
- Internet connection.

Installation

1. Check the computer for the CloudLink application (using a Siemens tool).
2. The **CloudLink.msi** file, when performed, simplifies CloudLink installation and configuration by displaying simple-to-read select and click options.

⇒ The application is installed at:
C:\Program Files\Siemens\CloudLink\[Version-number]

⇒ The CloudLink  icon displays in the task bar.

⇒ The **Siemens CloudLink** application is visible in the **Windows > Settings > Apps > Apps & features**.



Network security

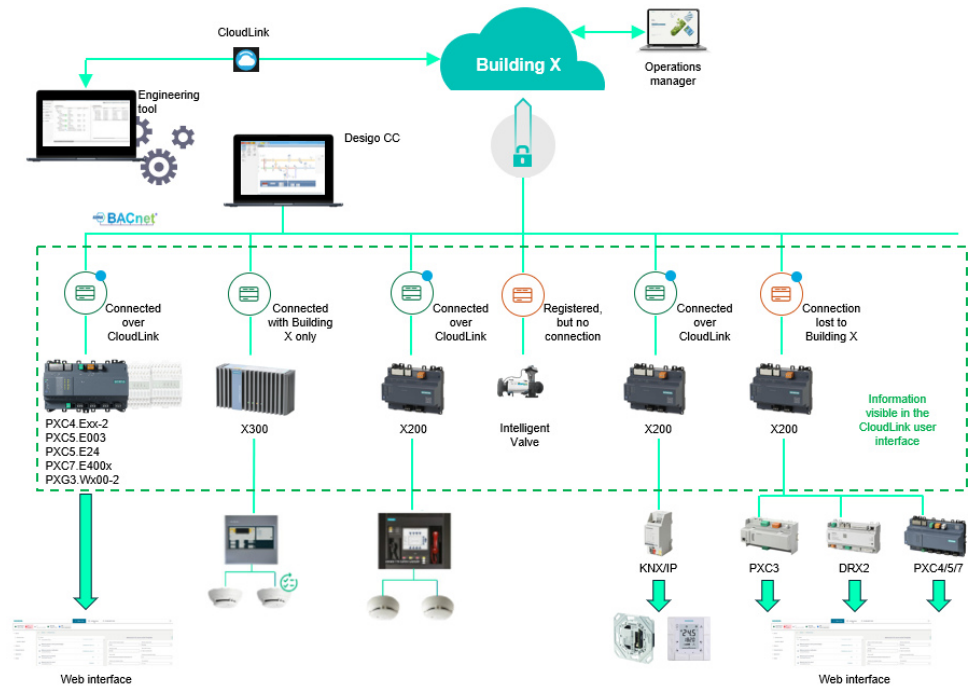
All security measures reduce potential gaps in network security or security exploitation vulnerabilities in:

- CloudLink
- X200
- X300
- Desigo
 - ABT Site
 - ABT Pro
 - ABT Go
 - Desigo PXC4/5/7
 - PXG3

⚠ WARNING**System manipulation due to unauthorized access**

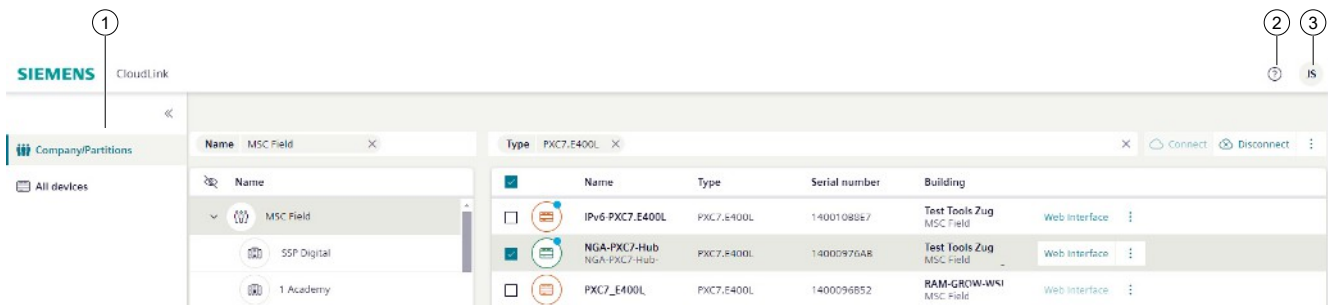
- Comply with the following items regarding security loopholes, vulnerabilities, and necessary security measures.
- Some system components do not meet current IT security expectations without implementing additional protection measures.
- Verify that your current Windows version is up to date (check for updates).
- The devices are not sufficiently protected against state-of-the-art IT attack strategies without suitable security measures and are at risk of exploitation through potential security loopholes or security vulnerabilities.
- A comprehensive risk assessment found that suitable and reasonable security measures can achieve appropriate and sufficient IT security on devices.
- Comply with the instructions in:
Connect Gateway Cybersecurity Guidelines [A6V12604901](#)
Building X Cybersecurity Guidelines [A6V13199528](#)
Desigo Cybersecurity Guidelines [A6V14142297](#)
Desigo PXC4/5/7 Cybersecurity Guidelines [A6V13813955](#).

3 CloudLink



The CloudLink software directly accesses connected device over the Building X environment on a local computer. The connection then provides access to the devices listed below, including smoke detectors, KNX/IP gateway, KNX end devices (for example, a room control unit). CloudLink can simultaneously connect and operate multiple devices.

Menus



	Menu	Submenu	Description
1	Vertical navigation		Lists devices sorted by Company/partitions or All devices .
2	Help menu		Accesses help and general information.
		Help	Opens the user guide from SID.
		Corporate Information	Displays information about the application.
		About	Displays the CloudLink version, Corporate information, Privacy Notice and Terms of use. The License area displays license information.

	Menu	Submenu	Description
3	User menu	Language	Selects the user interface language English or German.
		Theme	Switches between light and dark mode.
		Logout	Logs off CloudLink.

Icons and buttons

Icons	Description
	The device is onboarded, but not connected to Building X. Check why the device is no longer online.
	The device is registered and connected to Building X. The device sends information such as values, alarms, and trends and can download and upload from the engineering tool.
	The device is registered and connected to Building X and connected via CloudLink. The device sends information such as values, alarms, and trends and can download and upload from the engineering tool. Additional commissioning features are available and operational.
	The device was connected to Building X via CloudLink, but the connection between Building X and the device is lost or interrupted. The state takes some time to update, click Refresh to update the state.
	Company folder with descriptive name.
	Shows the devices in a flat list.
	Building with descriptive name.
 	Show or hide all nodes.
	▷ The device state must be connected . ◆ Select Connect and the device state changes to . ⇒ The device is connected over the CloudLink.

2. Is **Enable connectivity** set to **Yes**.
3. Is **Registration URL** defined as <https://bootstrap.connectivity.siemens.com>.

Infra'NwkPortCloud Properties

Enable connectivity Yes	Reconnect delay 01:00
Retry attempts 10	Activation key UBD5HY-ZY2XZ-ELMAR-
Connection status Connected	Last connected 25.06.2024 13:55:08
Connection URLs [0]	Registration URL https://bootstrap.connectivity.siemens.com

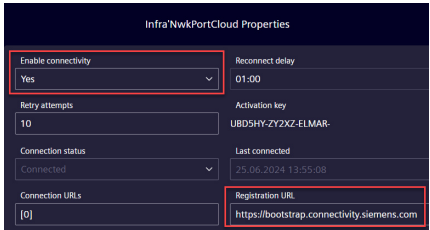
1. In the taskbar, right-click the CloudLink  and select **Open**.
2. Filter to select the desired online  device. Up to 10 devices can be connected at the same time.
3. Select **Connect**.

⇒ The device state changes to  and is operational.
4. Select **Web interface**.
⇒ The **Web interface** opens in a new browser window.

ETS specific

- ETS version > 5.4 is installed on the computer.
- For parallel downloads to multiple KNX lines: a KNXnet/IP interface can be specified in ETS for each KNX line.
- For several KNXnet/IP interfaces via a single X200 gateway: The CloudLink tool must provide several UDP/TCP connections via different ports, for example, 127.0.0.1:3671, 127.0.0.1:3672, ...
- Building X must be configured so that multiple connections (UDP, TCP, multicast) use only one connection from the CloudLink tool to the X200 at the customer site.

Troubleshooting

Icon	Possible cause	Task
	The device is not correctly onboarded in Building X.	Unregister the device and onboard the device again.
	The number of subscriptions ordered is not sufficient for this device.	Order additional subscriptions for this device.
	The subscription has not been renewed and is therefore no longer valid.	Renew the order for this device.
	The old device was replaced with another one and is no longer onboarded.	Unregister the device in Building X.
	No connection to the device available.	Check the internet connection to the device. Check the power supply to the device. Restart the port or reboot the computer. Desigo specific: Check that Cloud property Enable connectivity is Yes. Check that the Registration URL is https://bootstrap.connectivity.siemens.com 

Error messages or symptoms

Title	Description/symptom	Proposed solution
Multi-connection limit	One (or more) connections are already open, another one cannot be opened.	Either close the existing connections or increase the limit for the maximum number of parallel connections in appsettings.json (not recommended).
Missing TLS certificate	When opening the device user interface, a warning displays in the browser regarding an invalid certificate (either incorrect host/domain name or expired validity date).	CloudLink does not have a suitable (self-signed localhost) certificate available to intervene in the connection. Restart the CloudLink and accept both requests to generate/install a certificate.

Title	Description/symptom	Proposed solution
Proxy	Connection to Building X API is disrupted; login works, but no devices, sites, organizations can be loaded (in a scenario where the Internet connection must be established via a proxy).	Manually enter a proxy in appsettings.json that is used for communication with the Building X API.
Incorrect Building X environment	The list of organizations, sites and devices contains content other than what is expected.	CloudLink can switch between different Building X sites. It is probably set to the wrong instance in this case.
Fallback port	The device is connected, the device user interface can also be called up, but third-party applications cannot transfer data via the connection.	If a TCP or UDP port that is designated for the connection is already in use (either by another program or by a connection to another device), CloudLink selects a fallback port. However, the third-party application in question must also be able to take this into account.

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